

Nº7

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1. $\frac{dn}{dt} = (0.88 + 0.00008n(t))(N - n(t))$
2. $\frac{dn}{dt} = (0.00008 + 0.88n(t))(N - n(t))$
3. $\frac{dn}{dt} = (0.7t + 0.6\sin(t)n(t))(N - n(t))$

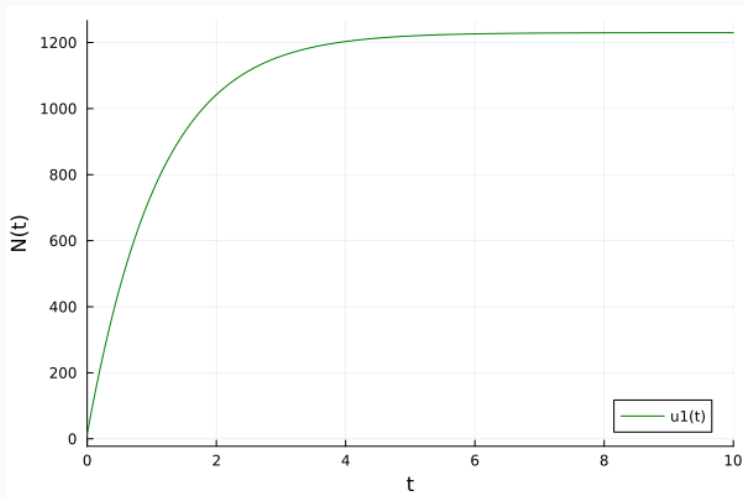
$$N = 1230,$$

$$14 \quad . \quad 2$$



```
using DifferentialEquations, Plots;
f(n, p, t) = (p[1] + p[2]*n)*(p[3] - n)
p1 = [0.88, 0.00008, 1230]
p2 = [0.00008, 0.88, 1230]
n_0 = 14
tspan1 = (0.0, 14.0)
tspan2 = (0.0, 0.02)
prob1 = ODEProblem(f, n_0, tspan1, p1)
prob2 = ODEProblem(f, n_0, tspan2, p2)
```

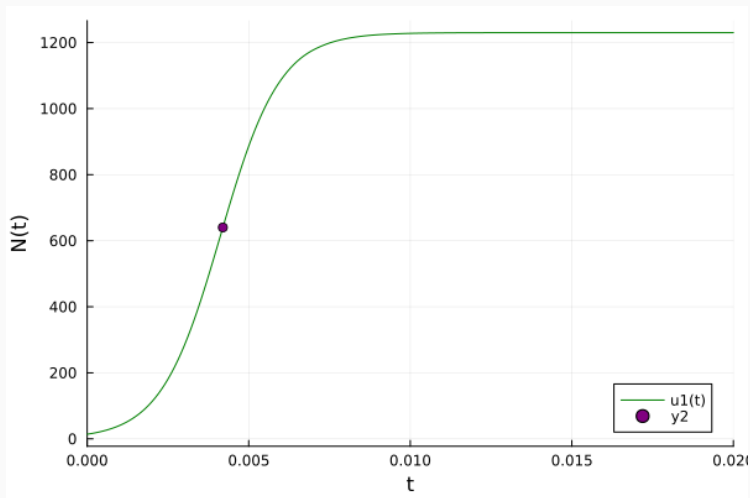
```
sol1 = solve(prob1, Tsit5(), saveat = 0.01)  
plot(sol1, markersize =:15, c =:green, yaxis = "N(t)")
```



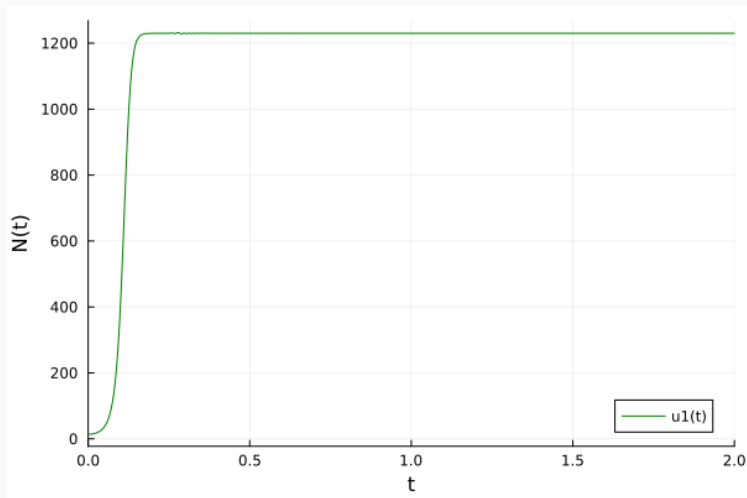
. 1:

1

```
sol2 = solve(prob2, Tsit5(), saveat = 0.0001)  
plot(sol2, markersize =:15, c=:green, yaxis="N(t)")
```



```
function f3(u,p,t)
    n = u
    dn = (0.7*t + 0.6*sin(t)*n)*(1230 - n)
end
u_0 = 14
tspan = (0.0, 2)
prob = ODEProblem(f3, u_0, tspan)
sol = DifferentialEquations.solve(prob, Tsit5(), saveat = 0.001)
plot(sol, markersize =:15, c=:green, yaxis="N(t)")
```

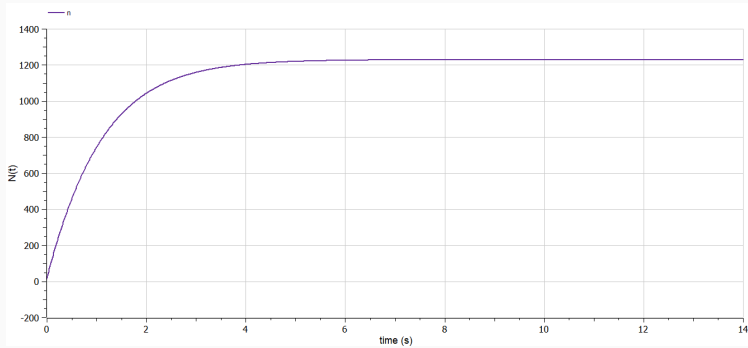


```
parameter Real a_1 = 0.88;  
parameter Real a_2 = 0.00008;  
parameter Real N = 1230;  
parameter Real n_0 = 14;
```

```
Real n(start=n_0);
```

```
equation
```

```
der(n) = (a_1 + a_2*n)*(N - n);
```



. 4:

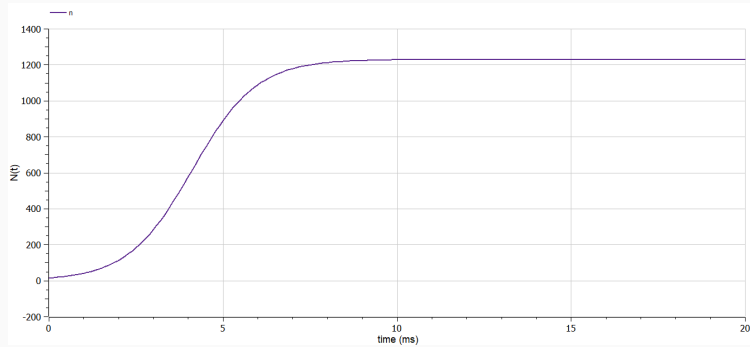
1

```
parameter Real a_1 = 0.00008;  
parameter Real a_2 = 0.88;  
parameter Real N = 1230;  
parameter Real n_0 = 14;
```

```
Real n(start=n_0);
```

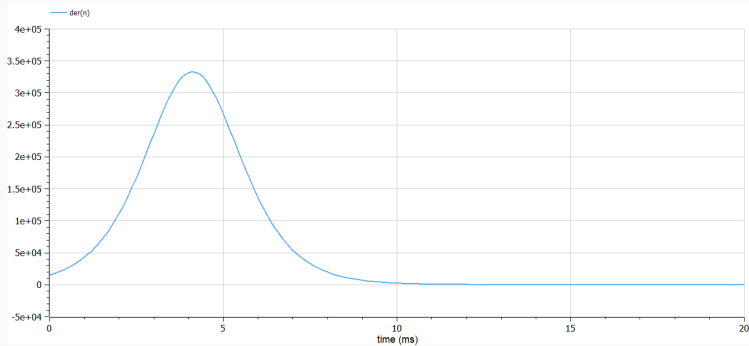
```
equation
```

```
der(n) = (a_1 + a_2*n)*(N - n);
```

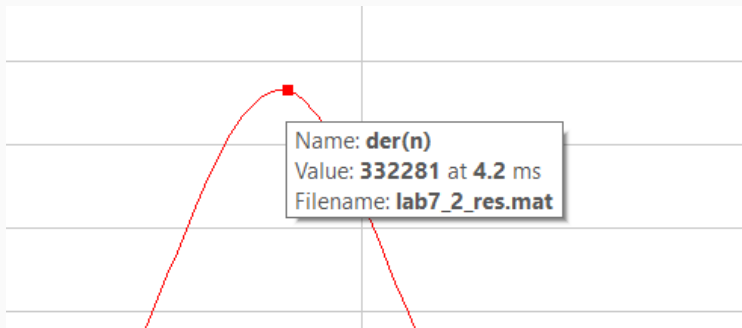


. 5:

2



. 6:



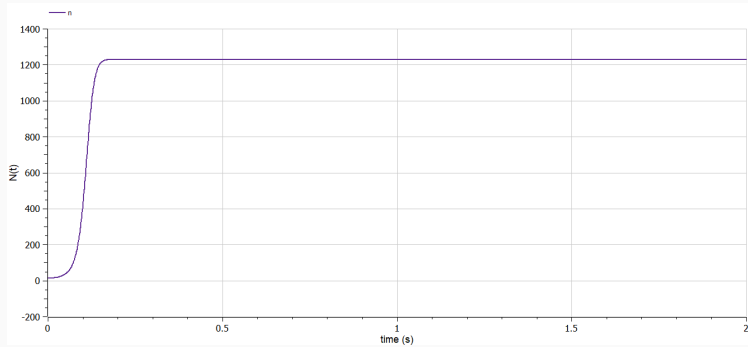
. 7:

```
parameter Real a_1 = 0.7;  
parameter Real a_2 = 0.6;  
parameter Real N = 1230;  
parameter Real n_0 = 14;
```

```
Real n(start=n_0);
```

```
equation
```

```
der(n) = (a_1*time + a_2*sin(time)*n)*(N - n);
```



. 8:

3



1. [].
URL:<https://studfile.net/preview/310591/>.